Epistract: A Two-Layer Knowledge Graph Framework with an Epistemic Super-Domain Layer

Comprehension-grounded extraction with categorical epistemic typing, validated across four pre-built domains

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Abstract

Most knowledge graph construction tools stop at extracting entities and typed relations: they tell you what a corpus says, but not *how* the corpus says it. A relation between a compound and a disease in a peer-reviewed clinical paper carries different epistemic weight from the same relation in a forward-looking patent claim, yet flat knowledge graphs collapse the distinction. I present Epistract, an open-source domain-agnostic knowledge graph framework distributed as a Claude Code plugin, that introduces a *two-layer* graph: Layer 1 contains the brute facts (entities, typed relations, evidence quotes, confidence scores), and Layer 2 is an *epistemic super-domain layer* that annotates every relation with a categorical status (`asserted`, `prophetic`, `hypothesized`, `contested`, `contradiction`, `negative`, `speculative`) using a rule engine combining regex patterns over evidence text with document-type inference. A single domain-specific analyst persona, defined as one YAML string, drives two complementary surfaces: a reactive chat interface over the graph, and a proactive auto-generated briefing that synthesizes the classified graph into temporally-stratified, contradiction-aware analyst output. The framework ships with four pre-built domains: drug-discovery (validated across six scenarios spanning neurogenetics, oncology, rare disease, immuno-oncology, cardiovascular inflammation, and GLP-1 competitive intelligence – 111 documents, 783 nodes, 2,230 relations across S1–S5 plus 278 nodes / 855 relations on S6 alone after a v3 rebuild), clinicaltrials (validated on a GLP-1 Phase 3 corpus from `ClinicalTrials.gov` – 10 protocols, 142 nodes, 395 relations), contracts, and fda-product-labels (the latter two are schema scaffolds awaiting their first showcase corpus). Each domain carries its own epistemic rules: phase-based evidence grading for clinicaltrials, a four-level FDA epistemology classifier (established / reported / theoretical / asserted) for fda-product-labels. Adding a new domain requires no pipeline changes – it is four configuration files (schema, extraction prompt, epistemic rules, persona) generated interactively via `/epistract:domain`. The framework, all corpora, and all artifacts are MIT-licensed and reproducible.[1]

This paper supersedes the March 2026 draft on GraphRAG-to-Epistract, which described the v1 framework as a single-domain (drug-discovery) system. The current paper documents the v3.2 framework: a two-layer architecture with a categorical epistemic super-domain layer, four pre-built domains with domain-specific epistemic rules, and a single-source-of-truth analyst persona pattern that drives both reactive chat and proactive briefings. The earlier paper is preserved at `paper/main.tex` *in the repository.*

[1] https://github.com/usathyan/epistract

1 Motivation: the missing dimension

Modern knowledge graph (KG) construction has converged on a recognizable pattern. Documents enter; an extractor (whether rule-based, fine-tuned NER, or LLM-driven) emits entities and typed relations; a graph builder deduplicates and resolves; a visualizer renders. The output is a labeled property graph that answers structural questions: which entities are most central, which clusters form, which relations are most prevalent. The flagship knowledge-graph platforms – Neo4j Bloom, Stardog, GraphDB, Microsoft's GraphRAG Edge et al. [2024], and a growing ecosystem of open-source LLM-based extractors – all deliver this primary product.

A flat knowledge graph, however, is fundamentally lossy with respect to the epistemic content of its source corpus. Consider three sentences that might appear in a drug-discovery corpus:

1. "In SURPASS-2, tirzepatide 15 mg reduced HbA1c by 2.30% versus 1.86% for semaglutide 1 mg (p<0.001)." Frías et al. [2021]

2. "In one or more embodiments, the compositions of the present invention may be useful for the treatment of cardiovascular disease, neurodegenerative disorders, and metabolic dysfunction."

3. "GLP-1 receptor agonism has been hypothesized to confer neuroprotective effects through mechanisms that are not yet fully characterized."

A flat KG would extract three relations of the form ⟨compound⟩ `INDICATED_FOR` ⟨disease⟩ from these sentences and present them as equivalent edges in the graph. Yet they are not equivalent. The first is an *asserted* relation backed by a randomized controlled trial. The second is *prophetic* – patent forward-looking language designed to maximize claim breadth, with no requirement that the indication be empirically established. The third is *hypothesized* – explicitly hedged research speculation. A drug-discovery analyst reading the graph cannot make decisions on the basis of edges that flatten this distinction.

This is not a problem that domain-specific extraction prompting solves. The information is present in the source text – the syntax of patents, hedging vocabulary in research papers, the structural conventions of trial reports – but a graph that records only ⟨ subject, predicate, object, confidence, source ⟩ has no field in which to store it. The missing dimension is *epistemic typing*: a categorical label, attached to each relation, that records what kind of claim it is.

This paper introduces Epistract, a domain-agnostic knowledge graph framework whose primary contribution is to treat the epistemic dimension as a first-class architectural layer rather than as ad-hoc per-relation metadata. Epistract produces *two-layer* graphs: Layer 1 is the conventional brute-facts graph (entities, typed relations, evidence quotes, confidence scores), and Layer 2 is an *epistemic super-domain layer* that annotates every relation with a categorical status using a rule engine combining regex patterns over the verbatim evidence text with document-type inference. The two layers are produced in the same pipeline, persisted in the same artifact (`claims_layer.json`), and rendered together in the visualization. A separate analyst-persona stage consumes the classified graph (not the source text) to produce a proactive briefing – contradictions surfaced, prophetic claims separated from asserted ones, follow-up actions recommended.

Two design choices distinguish the framework from prior work. The first is *domain pluggability*: the pipeline is domain-agnostic, and a domain is defined by four small configuration files (schema in YAML, extraction prompt in Markdown, epistemic rules in Python, analyst persona in YAML). The framework ships with four pre-built domains – drug-discovery, contracts, clinicaltrials, and fda-product-labels – and adding a new one requires no Python changes to the core pipeline. The second is the *persona dual-use pattern*: a single YAML string per domain (the analyst persona) is the system prompt for both a reactive chat surface (workbench, where users

ask ad-hoc questions) and a proactive briefing surface (the auto-generated narrator output). The two surfaces stay coherent because they derive from the same persona.

This paper is the second in a sequence. The first Bhatt [2026] described the v1 framework – a single-domain (drug-discovery) system validated against six scenarios – as a refutation of similarity-based GraphRAG Bhatt [2025], Edge et al. [2024] for biomedical literature analysis. The current paper documents the framework's evolution into v3.2: a two-layer architecture, four pre-built domains, the persona dual-use pattern, and the seventh validation scenario (clinical-trials). The v1 paper's argument – that LLM *comprehension* produces typed, evidence-backed relations that embedding-similarity cannot – remains intact and is the foundation Layer 1 still rests on. What is new is Layer 2 and the framework around it.

The remainder of this paper is organized as follows. Section 2 describes the two-layer architecture and the pipeline. Section 3 details the epistemic super-domain layer – the rule engine, the status taxonomy, and the persona-driven briefing. Section 4 describes the four pre-built domains and the `/epistract:domain` wizard for adding new ones. Section 5 reports validation across all seven scenarios spanning two active domains. Section 6 compresses the GraphRAG → Epistract v1 → Epistract v3 trajectory. Sections 7–9 discuss collaboration, availability, and limitations.

2 Architecture: a two-layer pipeline

Epistract implements a six-stage pipeline (Figure 1). Documents enter; an agent-based extractor produces typed entities and relations against a domain schema; a deterministic validation step verifies molecular identifiers (drug-discovery only); a graph builder deduplicates and detects communities; an *epistemic classifier* annotates every relation with a categorical status; and an analyst-persona stage produces a proactive briefing from the classified graph. The visualizer surfaces both layers together, with an interactive chat panel grounded in the same domain persona.

2.1 Agent-based extraction

The central extraction choice is unchanged from v1: extraction agents run as Claude Code subprocesses, not stateless API calls Bhatt [2026]. Each agent receives a full document, reads it with domain understanding informed by the domain's `SKILL.md` extraction prompt, and produces structured JSON conforming to the domain schema's entity and relation types. For corpora of four or more documents, agents execute in parallel, with one extractor per document. Each agent emits, per relation, a verbatim evidence quote and a confidence score in $[0, 1]$ – two fields the epistemic classifier in Layer 2 depends on.

This is qualitatively different from chunked-RAG retrieval and from API-based stateless extraction. A chunked retriever loses the cross-paragraph context that distinguishes a method statement from a result; a stateless API extractor, processing chunks in isolation, cannot resolve a phrase like "the same compound" across paragraph boundaries. By giving each agent the entire document and a long context window, Epistract preserves the discourse structure necessary for relation typing.

2.2 Domain pluggability

Domain knowledge enters the pipeline through four files, all in a self-contained per-domain directory:

- `domain.yaml` – entity types and relation types in YAML, with names, descriptions, and example surface forms. The *schema*.

- `SKILL.md` – the extraction prompt the agent reads. Includes naming conventions (INN names for drugs, HGNC symbols for genes, MeSH preferred terms for diseases), confidence-scoring guidance, and per-entity-type examples.
- `epistemic.py` – the per-domain rule engine that produces categorical statuses. Includes `HEDGING_PATTERNS` (regex over evidence text), `infer_doc_type()` (inference rules from filename or text features), and any domain-specific custom rules (e.g., the four-level FDA epistemology classifier in `fda-product-labels/epistemic.py`).
- `workbench/template.yaml` – the analyst persona, in one YAML field as a multi-line string. This is the system prompt for both the reactive chat and the proactive narrator.

The core pipeline (`core/`) is domain-agnostic. `core/domain_resolver.py` loads a domain by name, locates the four files, and threads the schema, prompt, and rule module through the rest of the pipeline. Adding a domain is a purely declarative act: write the four files, place them under `domains/<name>/`, and `/epistract:ingest --domain <name>` works. The framework ships with a `/epistract:domain` wizard that bootstraps these four files from a sample corpus through a guided three-pass interactive flow.

2.3 Deterministic validation (drug-discovery only)

Drug-discovery alone among the four shipped domains carries a deterministic post-extraction validation step. SMILES strings are validated and canonicalized through RDKit Landrum [2024]; protein and nucleotide sequences are checked against standard alphabets through Biopython Cock et al. [2009]; structured identifiers (CAS registry numbers, NCT trial numbers, InChIKeys) are validated by regex and, where possible, cross-referenced. The validation report is auto-emitted as `validation_report.json` alongside the graph. The other three shipped domains do not require this stage because their entity types are not chemistry-bound; the framework simply skips the validation step when the domain provides no `validation-scripts/` directory.

2.4 Graph construction

Once extraction and (optional) validation are complete, sift-kg Ceresa [2024] handles graph assembly: entity deduplication via SemHash and Unicode normalization, Louvain community detection, and interactive HTML visualization via pyvis Perrone [2018]. A custom labeling engine assigns descriptive community names from member entity composition rather than numeric identifiers (this remains the same as v1, with refinements). The output of this stage is `graph_data.json` – the canonical Layer 1 artifact.

2.5 Epistemic classification (Layer 2)

The new architectural element in v3 is a separate post-graph stage that annotates Layer 1 with categorical epistemic statuses. This is the subject of Section 3; in summary, the stage reads `graph_data.json`, applies the domain's rule engine (`epistemic.py`) to every relation's verbatim evidence and source-document type, and writes `claims_layer.json`. This artifact is the canonical Layer 2 output. It is reproducible (rule-based) and bit-identical across runs given identical Layer 1 input.

2.6 Persona-driven narration

A second post-graph stage consumes the classified graph (*not* the source documents) and produces a proactive briefing. The narrator's system prompt is the domain's analyst persona, loaded directly from `workbench/template.yaml`. Its input is a structured summary of the

classified graph – per-status counts, contested-claim pairs, prophetic-claim concentrations, gap signals – assembled by `core/label_epistemic.py` from `claims_layer.json`. The output is `epistemic_narrative.md`, a domain-shaped analyst briefing with executive summary, contested claims, prophetic claims, gaps, and recommended follow-ups. Unlike Layer 2, this output is non-deterministic (LLM-generated); the deterministic record is `claims_layer.json`.

2.7 Workbench: the reactive surface

A web-based dashboard (the workbench, launched via `/epistract:dashboard`) renders the graph with a chat panel beside it. The chat's system prompt is the same analyst persona used for the narrator. Both surfaces – the auto-generated briefing and the chat – are grounded in the same graph data and the same persona. This is the persona dual-use pattern: one source of truth (`template.yaml`'s persona field), two surfaces (proactive narrator, reactive chat). Section 3 returns to this point.

2.8 Plugin delivery

Epistract is distributed as a Claude Code plugin Anthropic [2024]. Users install with `/plugin install epistract` and invoke pipeline stages via slash commands: `/epistract:ingest`, `/epistract:epistemic`, `/epistract:dashboard`. The plugin encapsulates the entire dependency tree (Python virtualenv, sift-kg, RDKit, Biopython, optional sentence-transformers) and exposes a single entry point per pipeline stage. The same delivery model from v1 carries forward unchanged.

3 The Epistemic Super-Domain Layer

The epistemic super-domain layer is the central architectural contribution of v3. This section describes its rationale, taxonomy, rule engine, output artifact, and the persona-driven narrator stage that consumes it.

3.1 Rationale

A confidence score in $[0, 1]$ on a relation is a one-dimensional signal: it conveys how strongly the extractor believes a given subject–predicate–object triple is in the source text. It does not convey what *kind of claim* the source text is making. The same edge in the graph could be supported by source language that asserts ($p < 0.001$ in a randomized trial), prophesies ("in one or more embodiments…" in a patent), hypothesizes ("has been suggested to…" in a discussion section), or denies ("did not show a statistically significant…"). These differ *categorically*, not by degree, and a scalar confidence flattens them all into the same numerical surface.

The epistemic super-domain layer adds a second dimension. Every relation in the graph receives a categorical *epistemic status* from a small fixed taxonomy. The taxonomy was chosen to be (a) source-language-driven (every status corresponds to identifiable patterns in the verbatim evidence text), (b) decision-relevant (analysts make different decisions about `prophetic` vs. `asserted` edges), and (c) cross-domain general (the same taxonomy applies in drug-discovery, contracts, clinicaltrials, and FDA labels, with per-domain variations).

3.2 The status taxonomy

The shared epistemic status taxonomy contains seven categorical labels (Table 1). Six are positive classifications (a relation has been classified as one of these); the seventh, `unclassified`, is assigned when no rule matches. Domain-specific epistemic modules may add or refine statuses

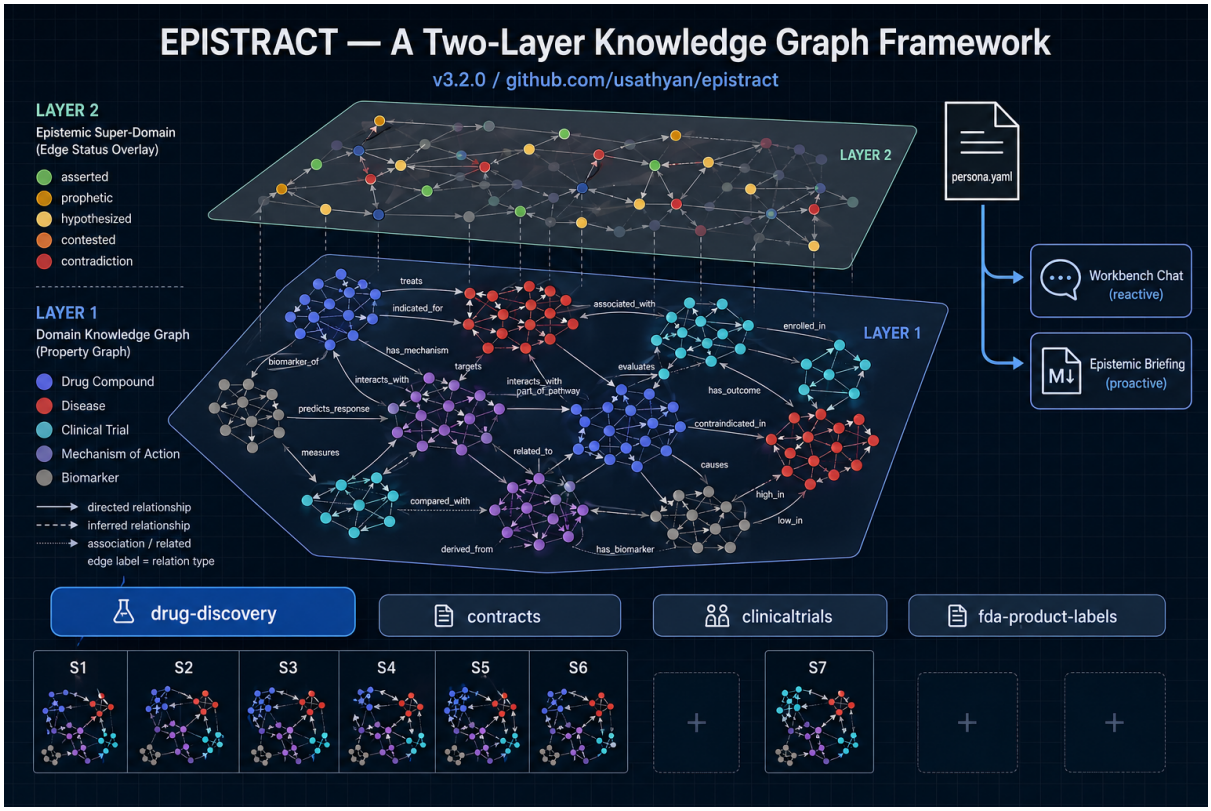

Figure 1: Epistract on one page. **Layer 1, the brute facts**: a force-directed labeled property graph with entities (compounds, diseases, mechanisms, biomarkers, clinical trials) extracted from a corpus and color-coded by entity type, connected by typed directional relations. **Layer 2, the epistemic super-domain**: a categorical status (`asserted`, `prophetic`, `hypothesized`, `contested`, `contradiction`, `negative`) annotated on every relation by a rule engine that combines regex patterns over evidence text with document-type inference. **The analyst persona** (right): a single `persona.yaml` string per domain drives both the reactive workbench chat and a proactive auto-generated briefing (`epistemic_narrative.md`) – one source of truth, two surfaces. **The four pre-built domains** (bottom): drug-discovery, contracts, clinicaltrials, fda-product-labels. **Track record** (S1–S7): scenario corpora that have pressure-tested each domain.

(for example, `fda-product-labels` introduces a four-level classifier mapping onto these statuses with finer granularity).

3.3 The rule engine

Layer 2 is produced by a deterministic rule engine in `core/label_epistemic.py`, parameterized by the domain's `epistemic.py` module. The engine has three stages, applied per relation:

1. **Document-type inference.** The source document's filename and any cached document-type hint (e.g., "patent_WO2024_*_zealand.txt" triggers patent-type inference) drive a coarse classification. Patent-source relations default to `prophetic` unless a more specific trigger fires.

2. **Hedging-pattern matching.** A list of compiled regex patterns runs against the relation's verbatim evidence text. Each pattern carries a status it produces on match. Patterns are short, conservative, and audited: they prefer `unclassified` over a wrong classification.

Table 1: The shared epistemic status taxonomy.

Status	Meaning and trigger patterns
`asserted`	Quantitative or definitive claim grounded in the source. Default for evidence text containing trial results, p-values, dose-response data, or unhedged assertion.
`prophetic`	Patent forward-looking language. Triggered by phrases such as "in one or more embodiments," "may be useful for," "the present invention provides," and document-type inference (the source document is a patent filing).
`hypothesized`	Hedged research language. Triggered by "hypothesized," "has been suggested," "may indicate," "is thought to," and similar epistemic hedging vocabulary.
`contested`	The same subject–predicate–object triple appears across multiple source documents with materially different confidence scores or contradictory evidence quotes. Detected at cross-source aggregation time, not within a single document.
`contradiction`	Explicit denial of a claim asserted in another source ("did not demonstrate," "failed to show statistical significance," "no evidence of…"). Stronger than `contested`.
`negative`	Explicit absence claim from a single source ("no association observed," "no detectable expression"). Distinct from `contradiction` (which is cross-source).
`speculative`	Marked future-conditional language outside patents ("further studies are needed," "the implications remain to be determined").
`unclassified`	No rule fired; default fall-through.

3. **Cross-source aggregation.** After per-relation classification, the engine groups relations by ⟨subject, predicate, object⟩ identity and inspects the multi-source set. If the same triple appears with confidence ranges spanning more than 0.30 across sources, it is annotated `contested`. If a source's evidence text is a denial of another source's assertion, the deniers are annotated `contradiction`.

The output of the rule engine is `claims_layer.json`: a per-relation record with the assigned status, the matching rule (for audit), the verbatim evidence quote, and a back-reference to the source document. This artifact is bit-identical across runs given identical Layer 1 input. It is the reproducible Layer 2 record.

3.4 Domain-specific epistemic rules

Each domain's `epistemic.py` can add domain-specific rules beyond the shared taxonomy. Two examples illustrate the pattern.

Phase-based evidence grading (clinicaltrials). Clinical trial documents from `ClinicalTrials.gov` carry trial-phase metadata (Phase 1, 2, 3, 4). The clinicaltrials epistemic module promotes Phase 3 and Phase 4 outcome relations to `asserted`, leaves Phase 2 at the rule-engine default, and demotes Phase 1 outcomes to `hypothesized` unless the evidence quote explicitly asserts a quantitative result. This phase-based grading captures regulatory-grade evidence stratification that a domain-agnostic rule engine cannot infer from text alone.

Four-level FDA epistemology (fda-product-labels). FDA Structured Product Labels carry a recognizable epistemic vocabulary ("has been demonstrated," "has been reported," "has

been theorized," implicit assertions in the Indications and Usage section). The fda-product-labels epistemic module implements a four-level classifier mapping onto the shared taxonomy with finer granularity: `established` (regulatory-grade asserted), `reported` (lower-tier asserted), `theoretical` (hypothesized in the FDA-specific sense), and a default `asserted` for unmarked indications-and-usage statements. The four levels are visible to the analyst persona, which can stratify briefings accordingly.

These per-domain extensions illustrate the framework's pluggability: the core pipeline does not need to know about phase-based grading or FDA epistemology vocabulary – the domain's `epistemic.py` encodes that knowledge, and `core/label_epistemic.py` dispatches to it.

3.5 The persona-driven narrator

A separate stage consumes the classified graph and produces a proactive briefing in natural language. The narrator's system prompt is the domain's analyst persona, loaded directly from `workbench/template.yaml`. The input to the narrator is *not* the source documents – it is a structured summary of the classified graph, assembled from `claims_layer.json`: per-status counts, contested-claim pairs with their confidence ranges, prophetic-claim concentrations by entity type, gap signals (entity types with low connection density), and any domain-specific signals the persona is configured to attend to. The narrator emits `epistemic_narrative.md`, structured into Executive Summary, Asserted Claims, Prophetic Claims, Contested Claims, Gaps, and Recommended Follow-Ups sections.

A worked example: on the S6 GLP-1 corpus (drug-discovery domain, 34 documents), the narrator's executive summary identified *61 prophetic claims* concentrated on cardiovascular, neurodegenerative, and metabolic-sub-disorder indications and recommended "temporal stratification of confidence" on the contested `semaglutide INDICATED_FOR obesity` edge (confidence range 0.55–0.97 across sources, where 0.55 instances trace to pre-STEP-1 patent language and 0.97 instances trace to post-approval clinical assertions). The narrator did not see the source documents directly; its claims trace back to graph nodes and relations annotated by the rule engine.

3.6 Persona dual-use: one source, two surfaces

The same persona string is also the system prompt for the workbench's reactive chat panel. When a user asks "which prophetic claims are concentrated on cardiovascular outcomes?" in chat, the answer is generated by the same persona that wrote the briefing, against the same classified graph, with the same vocabulary and decision criteria. This is by design: the proactive briefing and the reactive chat would otherwise drift apart in tone, vocabulary, and analytic framing if they were configured separately. The pattern reduces the surface area of the framework – one YAML field defines the domain's analyst voice – and makes domain extension cheaper: a domain author writes one persona, and both surfaces inherit it.

4 Pre-built Domains

The framework ships with four pre-built domains. Two have been validated against showcase corpora; two are schema scaffolds awaiting their first showcase corpus. Table 2 summarizes their state.

4.1 drug-discovery

The original Epistract domain Bhatt [2026]. Thirteen entity types: `COMPOUND`, `PROTEIN`, `DISEASE`, `MECHANISM_OF_ACTION`, `ORGANIZATION`, `PUBLICATION`, `CLINICAL_TRIAL`, `BIOMARKER`, `PHENOTYPE`, `ADVERSE_EVENT`, `PATHWAY`, `REGULATORY_ACTION`, `DOCUMENT`. Twenty-two directed relation types.

Table 2: The four pre-built domains in v3.2.0. "Scenarios validated" is the maturity signal – a domain that has been pressure-tested across multiple varied corpora has had its persona, epistemic rules, and entity-type coverage refined in ways one with zero scenarios has not. Schema is static.

Domain	Entities	Relations	Scenarios	Specialty pipeline
drug-discovery	13	22	S1–S6 (six)	RDKit / Biopython molecular validation; structural-node enrichment
contracts	11	11	none yet	—
clinicaltrials	12	10	S7 (one)	Phase-based evidence grading; optional CT.gov enrichment
fda-product-labels	17	16	none yet	Four-level FDA epistemology classifier

The schema was refined relative to v1 (which had 17 entities and 30 relations) by consolidating low-yield types – entity types and relations that, across S1–S5, were under-exercised or overlapped with existing types. The S6 V3 rebuild Bhatt [2026] validated the refined schema against a 34-document multi-source GLP-1 corpus.

The domain ships with deterministic molecular validation: SMILES strings are validated and canonicalized with RDKit, protein and nucleotide sequences with Biopython, and structured identifiers (CAS registry numbers, NCT numbers, InChIKeys) by regex. Validated identifiers become first-class graph nodes (`CHEMICAL_STRUCTURE`, `PEPTIDE_SEQUENCE`) linked to their parent entities, carrying computed properties as attributes (molecular weight, Lipinski compliance, GC content, isoelectric point). The validation report (`validation_report.json`) is auto-emitted alongside the graph.

The persona is a senior drug-discovery competitive-intelligence analyst, with vocabulary calibrated to therapeutic-area, mechanism-of-action, and regulatory framing. The same persona drives the workbench chat and the narrator briefing on every drug-discovery scenario.

4.2 clinicaltrials

A trial-protocol-aware domain Davidson and Bhatt [2026] for ingestion of `ClinicalTrials.gov` protocol records. Twelve entity types: `Trial`, `TrialPhase`, `Sponsor`, `Compound`, `Intervention`, `Condition`, `Cohort`, `Population`, `Outcome`, `Biomarker`, `Endpoint`, `DOCUMENT`. Ten directed relation types capturing trial structure (`HAS_PHASE`, `SPONSORED_BY`, `TESTS_INTERVENTION`, `ENROLLS_POPULATION`, `MEASURES_OUTCOME`, etc.).

The domain's epistemic module implements phase-based evidence grading: Phase 3 and Phase 4 outcome relations are promoted to `asserted`; Phase 2 retains the rule-engine default; Phase 1 outcomes are demoted to `hypothesized` unless the evidence quote contains explicit quantitative results. This captures a regulatory-grade evidence stratification that a domain-agnostic rule engine cannot infer from text alone.

An optional `--enrich` flag during ingestion fetches additional structured fields from the CT.gov API (NCT-keyed lookup, currently using `urllib` due to a `httpx` 403 issue with the CT.gov upstream that the framework documents in `docs/known-limitations.md`). Enrichment is opt-in because it adds API latency and is unnecessary for protocols that are already self-contained in their JSON exports.

The persona is a clinical trial design and competitive intelligence analyst, with vocabulary attuned to phase, endpoint hierarchy, sponsor strategy, and regulatory pathway.

4.3 contracts

A vendor- and event-contract domain. Eleven entity types covering contract parties (`Party`, `Vendor`, `Sponsor`), structural elements (`Obligation`, `Deadline`, `Deliverable`, `Payment`, `Termination`, `Renewal`), and metadata (`Document`, `Section`). Eleven directed relations capturing contractual structure. The domain has no validated showcase corpus; its inclusion in v3.2.0 demonstrates that the framework supports non-biomedical domains without pipeline modification.

4.4 fda-product-labels

A FDA Structured Product Label (SPL) domain. Seventeen entity types covering drug products (`DrugProduct`, `ActiveIngredient`, `Strength`, `DosageForm`), regulatory metadata (`NDC`, `NDA`, `ANDA`, `SetID`, `RxCUI`, `UNII`), label sections (`Indication`, `Contraindication`, `AdverseReaction`, `Interaction`, `Warning`, `LabTest`), and `DOCUMENT`. Sixteen directed relation types.

The domain's epistemic module implements a four-level FDA epistemology classifier (`established`, `reported`, `theoretical`, `asserted`) keyed to FDA's recognizable label-text vocabulary ("has been demonstrated," "has been reported," "has been theorized," unmarked Indications-and-Usage assertions). The four levels map onto the shared epistemic taxonomy with finer granularity, allowing the analyst persona to stratify briefings into "regulatory-grade evidence," "post-marketing reports," "mechanistic theory," and "unmarked indications."

The domain has no validated showcase corpus as of v3.2.0; the schema and epistemic module are scaffolds, awaiting a 200–400-document SPL ingestion that GitHub Issue #14 tracks. The persona has been hand-tailored to FDA-canonical citations (SPL set ID, NDA/ANDA, NDC, RxCUI, UNII) in anticipation of the showcase corpus.

4.5 Adding a new domain

A new domain is four files. The framework ships an interactive wizard, `/epistract:domain create <name> <sample-corpus-path>`, that bootstraps the four files in three passes:

1. *Sample analysis.* The wizard reads 5–10 documents from the sample corpus, identifies recurring entity surface forms, and proposes an initial schema (entity types and relation types).

2. *Schema refinement.* The user edits the proposed schema, accepts or rejects entity types, and approves naming standards. The wizard regenerates `domain.yaml`.

3. *Prompt and persona generation.* The wizard generates `SKILL.md` (extraction prompt with examples drawn from the sample corpus), `epistemic.py` (a starter rule engine with shared-taxonomy patterns plus any domain-specific additions the user requested), and `workbench/template.yaml` (a starter persona inferred from the domain's terminology).

The total wall-clock time for a new domain, with a sample corpus in hand, is typically under 15 minutes. The output is immediately runnable – `/epistract:ingest <corpus> --domain <new-name>` works on the freshly generated configuration.

5 Evaluation

The framework has been validated across seven scenarios spanning two of the four pre-built domains. This section presents per-scenario quantitative results, schema coverage analysis, epistemic-layer outputs, and an honest accounting of what the evaluation does and does not establish.

Table 3: Validation scenarios across active domains. "UATs" is the user-acceptance-test pass rate for that scenario. S6 numbers reflect the v3 rebuild against the refined drug-discovery schema (13 entities / 22 relations); S1–S5 numbers retain the v1 schema (17 entities / 30 relations) and are reported as published in the v1 paper Bhatt [2026].

#	Domain / Corpus	Docs	Nodes	Relations	Comm.	UATs	Schema ver.
S1	drug-discovery / Neurogenetics (PICALM)	15	149	457	6	4/4	v1
S2	drug-discovery / Oncology (KRAS G12C)	16	108	307	4	5/5	v1
S3	drug-discovery / Rare Disease	15	94	229	4	3/3	v1
S4	drug-discovery / Immuno-Oncology	16	132	361	5	4/4	v1
S5	drug-discovery / CV & Inflammation	15	94	246	5	3/3	v1
S6	drug-discovery / GLP-1 CI (V3 rebuild)	34	278	855	10	6/6	v3
S7	clinicaltrials / GLP-1 Phase 3 (CT.gov)	10	142	395	4	5/5	v3.1
	Drug-discovery (S1–S6)	111	855	2,455	34	25/25	
	Clinicaltrials (S7)	10	142	395	4	5/5	

5.1 Scenarios across domains

Table 3 reports the seven validation scenarios. The first six (S1–S6) exercise the drug-discovery domain across varied therapeutic areas; the seventh (S7) is the inaugural clinicaltrials scenario.

The drug-discovery domain has accumulated the most pressure-testing. S1–S5 establish breadth across therapeutic areas; S6 added multi-source corpus assembly (PubMed E-utilities, Google Scholar, Google Patents via SerpAPI) and stress-tested patent-document handling. S7 is the first scenario in a second domain and exercises clinicaltrials' phase-based evidence grading and CT.gov ingestion. Contracts and fda-product-labels do not yet have validated scenarios; they are schema scaffolds.

5.2 Cross-source patent extraction (S6)

The S6 GLP-1 corpus comprises 24 PubMed abstracts and 10 patent filings from five pharmaceutical companies (Novo Nordisk, Eli Lilly, Pfizer, Hanmi Pharmaceutical, Zealand Pharma). The patent component is a deliberate stress test: patents differ structurally from research papers, use heavily prophetic language, and contain molecular identifiers (peptide sequences, CAS registry numbers, InChIKeys) that abstracts do not. The rebuild produced 278 entities and 855 relations on the refined v3 schema – a higher density than any S1–S5 corpus and the first scenario to exercise multi-source entity merging at scale (semaglutide appears in 17 documents – 4 patents and 13 papers – and is correctly deduplicated to a single node with merged attributes).

5.3 Trial-protocol structure (S7)

The S7 GLP-1 Phase 3 corpus comprises 10 trial protocols from `ClinicalTrials.gov`, hand-selected to span the contemporary GLP-1 receptor agonist clinical landscape (semaglutide STEP and SUSTAIN families, tirzepatide SURPASS family, and orforglipron). The clinicaltrials domain extracted 142 nodes and 395 relations, with the largest entity-type contributions from `Outcome` (65), `Intervention` (16), and `Cohort` (13). The phase-based grading promoted Phase 3 outcome relations to `asserted`; the resulting briefing surfaced regulatory-grade head-to-head evidence (e.g., SURPASS-2 tirzepatide-vs-semaglutide) distinct from earlier-phase signals.

5.4 Schema coverage analysis

Across the six drug-discovery scenarios under the v1 schema, the pipeline exercised 17 of 17 entity types (100%) and 28 of 30 relation types (93%); the two unused relations were `PHOSPHORYLATES` and `METABOLIZED_BY`, which represent specialized biochemical relationships

rarely discussed at the abstract level. The v3 schema refinement consolidated underused types: the refined 13-entity / 22-relation schema retained the high-yield types and merged or removed the low-yield ones. S6 V3 exercised all 13 entity types and 20 of 22 relation types (the two unused being domain-rare structural relations). On the clinicaltrials domain, S7 exercised 11 of 12 entity types (92%, the unused being `Endpoint` as a separate node from `Outcome`) and 9 of 10 relation types (90%).

5.5 Epistemic-layer outputs (S6 case study)

The S6 V3 `claims_layer.json` contained 855 relation classifications: ∼430 `asserted`, 61 `prophetic`, 78 `hypothesized`, 12 `contested` pairs, 2 `contradiction` pairs, 14 `negative` entries, and the remainder `unclassified` or `speculative`. The 61 prophetic claims concentrated on cardiovascular, neurodegenerative, and metabolic-sub-disorder indications – a pattern consistent with patent claim-breadth strategy in the GLP-1 receptor agonist class. The 12 contested pairs included high-stakes edges such as `semaglutide INDICATED_FOR obesity` (confidence range 0.55–0.97 across sources) where pre- and post-STEP-1 trial language coexists in the corpus.

The narrator's executive summary on S6, generated from the classified graph by the senior drug-discovery competitive-intelligence persona, reproduced the prophetic-concentration finding and recommended "temporal stratification of confidence" on the contested obesity-indication edge. It also surfaced an ingestion gap – absence of SURPASS-2 data in the corpus – and recommended that the analyst integrate the trial. The recommendation was correct: SURPASS-2 (Frías et al., NEJM 2021) Frías et al. [2021] was not in the original corpus and is the canonical head-to-head tirzepatide-vs-semaglutide evidence. The narrator did not see the source documents directly; its claims trace back to graph nodes and statuses set by the rule engine.

The narrator output is non-deterministic: re-running the persona-driven stage on the same `claims_layer.json` produces briefings that vary in word choice but stabilize on the same findings (prophetic concentrations, contested edges, gap signals). The deterministic record is `claims_layer.json`; the briefing is its narrative rendering.

5.6 What the evaluation establishes

The evaluation is sufficient to establish the following claims:

- *Domain pluggability is real, not aspirational.* A second active domain (clinicaltrials) ships in v3.1 with its own schema, epistemic rules, and persona, and validates with a first scenario. The pipeline did not require modification.
- *The rule-based epistemic layer produces analyst-relevant signal.* On S6, the layer correctly identified prophetic concentrations and contested edges that match the substantive structure of the GLP-1 corpus.
- *The persona dual-use pattern is coherent.* On both S6 and S7, the chat surface and the briefing surface use the same vocabulary and answer the same questions consistently because they share a persona string.
- *Scenario count is a maturity signal.* Drug-discovery (six scenarios) and clinicaltrials (one scenario) differ in pressure-testing depth, and that difference shows in subjective briefing quality and edge-case handling. The framework exposes this gap honestly rather than masking it.

5.7 What the evaluation does not establish

The evaluation does *not* establish:

- *Cross-scenario auto-learning.* The framework does not currently fold lessons from prior scenario runs into subsequent runs. Refinements (improved persona vocabulary, refined epistemic rules, schema consolidation) are human-mediated: a contributor reads the narrator output, identifies a missed pattern, edits the domain configuration, and commits. The README and the framework treat scenario count as a track record, not as evidence of compounding learning. GitHub Issue #15 tracks the aspirational mechanism.
- *Universal coverage of FDA labels or contracts.* Two of the four pre-built domains lack validated showcase corpora. Their schemas are designed but not pressure-tested. GitHub Issue #14 tracks the fda-product-labels showcase corpus.
- *Narrator output reliability for clinical decisions.* The narrator is a research tool, not a regulatory or clinical decision-making system. Its outputs are hypotheses requiring expert review, and the system documents this caveat both in-product and in the README.

These caveats are not afterthoughts. They are first-class limitations the framework documents explicitly and tracks in the issue queue. Section 8 expands on the responsible-use boundary.

6 Evolution: GraphRAG → Epistract v1 → Epistract v3

This section compresses the multi-version trajectory of the framework. Readers familiar with the v1 paper Bhatt [2026] can skim; readers new to the system may find the comparison useful as orientation.

From GraphRAG to v1: similarity to comprehension. The v1 paper argued that biomedical KG construction needs typed, directional, evidence-backed relations grounded in domain ontologies, and that LLM comprehension of full documents – not embedding similarity over chunks – is the right substrate. That argument remains the foundation of Layer 1 in v3.

From v1 to v3: comprehension is necessary but not sufficient. A graph of typed, evidence-backed relations is still flat with respect to the epistemic content of its sources. A patent's prophetic indication is recorded as the same kind of edge as a Phase 3 trial's asserted indication. The v3 architecture adds a second layer that records categorical epistemic statuses, and a persona-driven narrator that synthesizes the two layers into analyst-shaped briefings. The schema also evolved: drug-discovery's 17-entity / 30-relation v1 schema was consolidated to 13 / 22 in v3 by removing low-yield types observed across S1–S5.

From v1 to v3: from one domain to four. v1 was a single-domain framework that the paper described, slightly imprecisely, as "six domains" (the six scenarios were varied corpora within drug-discovery, not separate domains). v3 ships four genuinely separate domains, each with its own schema, epistemic rules, and persona, with the core pipeline domain-agnostic. The clinicaltrials domain in particular illustrates the value of per-domain epistemic rules: phase-based grading is incompatible with drug-discovery's hedging-and-doc-type model, but fits naturally as a domain extension.

From v1 to v3: from extraction artifact to analyst surface. v1 produced a graph; the user explored it manually. v3 adds two reading surfaces: a proactive briefing (`epistemic_narrative.md`) auto-generated per run, and a reactive chat (the workbench) grounded in the same persona and graph. Both are downstream of `claims_layer.json`, the new canonical artifact.

Table 4: Architectural evolution across three iterations.

Dimension	GraphRAG	Epistract v1	Epistract v3
Relationship basis	Embedding similarity	LLM comprehension	LLM comprehension + epistemic typing
Layers	One	One (brute facts)	Two (brute facts + epistemic super-domain)
Schema	Generic NER	17 ent / 30 rel, drug-discovery only	4 domains: 11–17 ent / 10–22 rel; refined drug-discovery 13/22
Domain pluggability	N/A	N/A	Four config files; `/epistract:domain` wizard
Relation typing	Untyped	Typed, directional	Typed + categorical epistemic status
Evidence	None	Source passage + confidence	+ per-relation epistemic status, classification rule, doc-type inference
Validation	None	RDKit, Biopython, NCT/CAS	Same; opt-in CT.gov enrichment for clinicaltrials
Briefing	None	None	Auto-generated narrator briefing per run
Chat surface	None	None	Workbench chat; same persona as briefing
Persona	N/A	N/A	One YAML string per domain; dual-use
Scale	6 articles	111 documents, 6 scenarios	121 documents, 7 scenarios, 2 active domains
Cross-domain	Single corpus	Six varied corpora, one domain	Two domains validated, two scaffolded
Source types	PubMed	PubMed + Scholar + Patents	+ ClinicalTrials.gov
Reproducibility	Code on GitHub	Plugin + committed artifacts	+ reproducible epistemic layer; non-deterministic narrator
Documented gaps	None enumerated	None enumerated	Issues #14 (fda showcase corpus), #15 (auto-learning), `docs/known-limitations.md`

Honest accounting. The v3 framework is also more explicit about what it does and does not do. The README, the in-product documentation, and this paper all distinguish "track record" (a domain has been pressure-tested across multiple scenarios) from "compounding learning" (the framework would automatically benefit from prior runs). The framework does the first; the second is aspirational and tracked as Issue #15. This shift in framing – from optimistic feature claims to documented gaps – is itself a contribution of v3.

7 Human-AI Collaboration Process

The v1 paper Bhatt [2026] described the collaboration model under which the framework was built: intensive interactive Claude Code sessions in which the human drove scientific strategy and acceptance criteria, and the AI drove implementation, parallelization, and documentation. That description remains accurate. This section adds three v3-specific observations.

7.1 The GSD workflow as methodology

The v3 development cycles were managed under a structured planning workflow ("Get Shit Done," GSD), in which each phase of work begins with a written goal and acceptance criteria, proceeds through a discussion-and-research stage, lands a plan, executes against atomic commits, and verifies goal achievement before closing. The workflow's artifacts (`ROADMAP.md`, `REQUIREMENTS.md`, per-phase `PLAN.md` and `VERIFICATION.md`) are local-only by design (gitignored under `.planning/`) but they shape the public commit history: each commit message names its phase and the requirement it discharges, producing an auditable trail from goal to implementation to verification.

This methodology made the multi-version trajectory of v3 tractable. Phase 18 ("Graph Fidelity & Honest Limits") landed FIDL-01 through FIDL-09 – nine independently verified requirements that together added the two-layer architecture, the persona dual-use pattern, and the v3 schema refinement. Each requirement was scoped tightly enough to verify in isolation; their composition produced the v3.0 release. Phase 21 (clinicaltrials) followed the same pattern for the second domain.

7.2 External contribution: the controlled-port pattern

The clinicaltrials domain (Section 4.2) and the fda-product-labels domain (Section 4) were initially contributed by Chris Davidson via pull request. In both cases the original contribution arrived against a stale fork (the contributor's main branch lagged the upstream main by a release or more), and a direct rebase or merge would either lose context or pull in unrelated drift. The framework adopted a *controlled-port pattern*: rather than rebasing the contributor's branch onto upstream main, the maintainer creates a new branch off current upstream main, hand-ports the substantive contribution file-by-file with attribution preserved via Git's `Co-Authored-By` trailer, and reviews each ported file against current conventions. This avoided two recurring failure modes (contributor-introduced `.planning/` files leaking into the public repo, and stub configurations replacing hand-tailored files) and preserved authorship attribution. The pattern is documented in repository memory for future contributions.

7.3 Honesty as a development practice

Several v3 commits are corrections rather than additions. The README was found to overclaim "domain knowledge compounds across scenarios – patterns generalize, edge cases surface, the analyst persona learns what to flag," implying an automatic-learning mechanism that the framework does not implement. The correction reframed scenario count as a track record (pressure-testing depth) rather than an auto-learning signal, and opened GitHub Issue #15 to track the aspirational mechanism. The visible commit (`a8a91db`) does not add a feature; it walks back a marketing claim. The collaboration model includes the human's role in catching such drifts – the AI alone is unlikely to identify when its own framing has shifted from "claim" to "promise."

The pattern recurs in three places: the README compounding correction, the demo-script accuracy checklist (every claim mapped to a verifiable codebase source), and this paper's Section 5 ("What the evaluation does not establish"). Each is an artifact of treating documentation honesty as a first-class development task rather than as polish.

8 Availability, Reproducibility, and Limitations

8.1 Availability

Epistract is released as open-source software under the MIT license at https://github.com/usathyan/epistract. Installation is two commands: `/plugin install epistract` (within

Claude Code) and `/epistract:setup` (which installs the Python pipeline, sift-kg, RDKit, and Biopython into an isolated virtual environment). The runtime requirements are Python 3.11 or later, Claude Code, and (for the chat and narrator stages) a credential for one of three supported LLM gateways: Azure AI Foundry, Anthropic direct, or OpenRouter, resolved in that priority order.

8.2 Reproducibility

All seven validation scenarios are reproducible from the public repository. Each scenario directory under `tests/corpora/` contains the source documents (or the SerpAPI / E-utilities query parameters used to assemble them, where licensing prohibits redistribution), the extraction outputs in JSON, the constructed graph data, the classified `claims_layer.json`, and the narrator's `epistemic_narrative.md`. The classified layer is bit-identical across runs given identical Layer 1 input. The narrator output is non-deterministic but stable in structure (executive summary, prophetic, contested, gaps, follow-ups) and findings (run-to-run word-choice variance, run-to-run finding-set stability).

The plugin architecture provides a versioning benefit: the deployed framework version is recorded in every output artifact's metadata, and committed test baselines serve as regression anchors across releases. The v3.2.0 release notes name the four shipped domains, the seven validated scenarios, and the gaps tracked in Issues #14 and #15.

8.3 Documented gaps

The framework documents three classes of gap explicitly:

Issue #14: fda-product-labels showcase corpus. The domain ships with schema, epistemic rules, and persona, but no validated showcase corpus. A 200–400-document SPL ingestion is in flight with the contributor.

Issue #15: cross-scenario knowledge persistence. The framework does not currently fold lessons from prior scenario runs into subsequent runs. Five candidate mechanisms for compounding knowledge across scenarios are scoped (cross-scenario entity canonicalization, domain-level aggregate graphs, persona few-shot from prior briefings, epistemic-rule mining, cross-scenario claim arbitration), with the first identified as the most tractable v3.3 candidate. This is aspirational; no work has begun.

`docs/known-limitations.md`. A running ledger of operational issues that constrain the framework: `httpx` 403 responses against `ClinicalTrials.gov` (use `urllib` instead), agent-tool model overrides failing silently when the parent runs a 1M-context Sonnet variant, workbench chat `max_tokens` truncation at 4096, and others. Each entry is logged with a workaround and a tracking reference.

8.4 Responsible use

Epistract is a research tool, not a clinical or regulatory decision-making system. The knowledge graphs it produces are derived from LLM extraction (probabilistic) and rule-based epistemic classification (deterministic, but rule-bounded). The narrator briefings are LLM-generated and non-deterministic. Every output should be treated as a hypothesis requiring expert review before informing any scientific, clinical, or regulatory decision. The framework explicitly discloses this caveat in the README, in the in-product workbench, and in this paper.

The persona pattern is a particular focus of the responsible-use boundary. A senior drug-discovery competitive-intelligence analyst is a *voice*; the framework does not certify that any

briefing meets professional standards, and users in pharmaceutical and healthcare settings should apply institutional review and domain expert oversight before acting on any framework output.

8.5 Community contribution

The framework's design invites contribution at four levels: new evaluation scenarios with curated corpora that pressure-test existing domains; schema extensions that introduce entity or relation types; new pre-built domains beyond the four shipped (the `/epistract:domain` wizard reduces the scaffolding cost to under 15 minutes for a new domain with a sample corpus); and external database connectors for entity enrichment (PubChem, ChEMBL, UniProt, ClinicalTrials.gov). Contribution patterns are documented in `docs/ADDING-DOMAINS.md`; the controlled-port pattern (Section 7) is documented for cross-version contributions.

9 Conclusion

The v1 framework demonstrated that LLM comprehension of full documents can produce typed, evidence-backed knowledge graphs that similarity-based assembly cannot. v3 extends that foundation in three architectural directions. First, a categorical *epistemic super-domain layer* annotates every relation with a status (`asserted`, `prophetic`, `hypothesized`, `contested`, `contradiction`, `negative`, `speculative`) using a deterministic rule engine that combines hedging-pattern matching with document-type inference. Second, the framework becomes genuinely *domain-agnostic*: a domain is four configuration files (schema, prompt, epistemic rules, persona), and the core pipeline does not need to be modified to add one. Third, the analyst persona is reified as a *single source of truth* per domain that drives both a proactive narrator briefing and a reactive workbench chat, keeping the two surfaces coherent.

The framework ships with four pre-built domains. Two – drug-discovery (six scenarios, 111 documents) and clinicaltrials (one scenario, 10 protocols) – are validated. Two – contracts and fda-product-labels – are schema scaffolds awaiting their first showcase corpus. The framework documents this gap honestly and tracks it in the public issue queue rather than presenting all four domains as if at parity.

The v3 contribution is methodological as much as architectural. Treating the epistemic dimension as a first-class layer changes what a knowledge graph is for: not a structural artifact a human reads in a visualizer, but the input to an analyst persona that synthesizes briefings and answers reactive questions. The persona dual-use pattern – one YAML string, two surfaces – collapses what would otherwise be two separately-maintained system prompts into one and prevents drift. The controlled-port pattern (Section 7) shows that the same architectural choices that make the framework pluggable also make external contribution tractable: a domain is a directory, not a patch against the core pipeline.

Several directions remain open for v3.3 and beyond. Cross-scenario knowledge persistence (Issue #15) would let a domain compound across runs – a canonical entity registry, then aggregate graphs, then learned rules – and is the most architecturally meaningful next step. The fda-product-labels showcase corpus (Issue #14) will exercise the four-level FDA epistemology classifier under real volume. Additional domains beyond the four shipped will test whether the wizard's 15-minute claim holds across diverse corpora. And the integration with regulatory and clinical workflows – the responsible-use boundary – needs continuing attention as the framework's outputs move from research artifacts to decision inputs.

The framework, all corpora, all artifacts, and this paper are reproducible from https://github.com/usathyan/epistract under the MIT license.

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